

संभव BATCH - CRASH COURSE

Mathematics: Average, Ratio & Proportion

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SSC CGL Level Exams | 50 Premium Practice Questions | No Answer Key

Q1. The average of the first 100 positive integers is:

- (A) 50
- (B) 50.5
- (C) 51
- (D) 100

SSC CGL Tier-I 2020

Q2. The average of 9 consecutive numbers is 37. What is the largest of these numbers?

- (A) 41
- (B) 43
- (C) 39
- (D) 45

SSC CGL Tier-II 2019

Q3. The average of 5 numbers is 27. If one number is excluded, the average becomes 25. The excluded number is:

- (A) 35
- (B) 25
- (C) 45
- (D) 30

SSC CHSL 2021

Q4. The average weight of 8 persons increases by 2.5 kg when a new person comes in place of one of them weighing 65 kg. The weight of the new person is:

- (A) 75 kg
- (B) 80 kg
- (C) 85 kg
- (D) 90 kg

SSC CPO 2020

Q5. A batsman makes a score of 87 runs in the 17th inning and thus increases his average by 3. Find his average after 17 innings.

- (A) 39
- (B) 38
- (C) 38.5
- (D) 39.5

SSC CGL Tier-I 2022

Q6. Out of 3 numbers, the first is twice the second and is half of the third. If the average of the three numbers is 56, the difference between the first and third number is:

- (A) 48
- (B) 24
- (C) 36
- (D) 72

SSC CGL 2021

Q7. The average of 11 results is 50. If the average of the first six results is 49 and that of the last six is 52, the sixth result is:

- (A) 56
- (B) 52
- (C) 50
- (D) 54

SSC MTS 2022

Q8. The mean of 50 observations was 36. It was found later that an observation 48 was wrongly taken as 23. The corrected new mean is:

- (A) 35.2
- (B) 36.1
- (C) 36.5
- (D) 37.1

SSC CHSL 2019

Q9. The average age of a husband and his wife was 23 years at the time of their marriage. After five years they have a one-year-old child. The average age of the family now is:

- (A) 19 years
- (B) 23 years
- (C) 18.5 years
- (D) 20 years

SSC CPO 2022

Q10. The average temperature of Monday, Tuesday, and Wednesday was 40°C . The average of Tuesday, Wednesday, and Thursday was 41°C . If the temperature on Thursday was 42°C , what was the temperature on Monday?

- (A) 38°C
- (B) 39°C
- (C) 40°C
- (D) 41°C

SSC Selection Post 2021

Q11. The average of the squares of first 10 natural numbers is:

- (A) 35.5
- (B) 36
- (C) 38.5
- (D) 40.5

SSC CGL Tier-I 2018

Q12. If the average of a , b , and c is M and $ab + bc + ca = 0$, then the average of a^2 , b^2 , and c^2 is:

- (A) M^2
- (B) $3M^2$
- (C) $6M^2$
- (D) $9M^2$

SSC CGL Tier-II 2020

Q13. In a class of 40 students, 45% are girls and the remaining are boys. If the average of the girls' marks is 54 and that of the boys is 46, what is the average of the whole class?

- (A) 49.6
- (B) 49.8
- (C) 49.5
- (D) 50.2

SSC CGL 2021

Q14. The average of 30 numbers is 40 and that of other 40 numbers is 30. The average of all the numbers is:

- (A) 34.28
- (B) 35.5
- (C) 34
- (D) 34.5

SSC MTS 2021

Q15. A library has an average of 510 visitors on Sundays and 240 on other days. The average number of visitors per day in a month of 30 days beginning with a Sunday is:

- (A) 285
- (B) 290
- (C) 295
- (D) 300

SSC CHSL 2020

Q16. If $x : y = 3 : 4$, then the value of $(4x + 5y) : (5x - 2y)$ is:

- (A) 7 : 32
- (B) 32 : 7
- (C) 11 : 4
- (D) 4 : 11

SSC CGL Tier-I 2022

Q17. The fourth proportional to 12, 21, 8 is:

- (A) 14
- (B) 16
- (C) 18
- (D) 20

SSC CPO 2021

Q18. The third proportional to 9 and 12 is:

- (A) 16
- (B) 15
- (C) 18
- (D) 21

SSC CGL 2020

Q19. The mean proportional between 0.08 and 0.18 is:

- (A) 0.12
- (B) 0.144
- (C) 0.012
- (D) 1.2

SSC MTS 2019

Q20. If $A : B = 3 : 4$ and $B : C = 8 : 9$, then $A : B : C$ is:

- (A) $3 : 8 : 9$
- (B) $6 : 8 : 9$
- (C) $3 : 4 : 9$
- (D) $8 : 6 : 9$

SSC CHSL 2022

Q21. If $2A = 3B = 4C$, then $A : B : C$ is:

- (A) $2 : 3 : 4$
- (B) $4 : 3 : 2$
- (C) $6 : 4 : 3$
- (D) $3 : 4 : 6$

SSC CGL Tier-I 2021

Q22. A sum of Rs 12,500 is distributed among A, B, and C such that A gets $\frac{2}{9}$ of what B and C together get. C gets $\frac{3}{4}$ of what A and B together get. The share of A is:

- (A) Rs 2250
- (B) Rs 2500
- (C) Rs 3000
- (D) Rs 2000

SSC CGL Tier-II 2019

Q23. The ratio of incomes of A and B is 5 : 4 and the ratio of their expenditures is 3 : 2. If at the end of the year, each saves Rs 1600, then the income of A is:

- (A) Rs 4000
- (B) Rs 3600
- (C) Rs 4500
- (D) Rs 5000

SSC CPO 2020

Q24. A bag contains 1 rupee, 50-paise, and 25-paise coins in the ratio 2 : 3 : 5. If the total value is Rs 114, then the number of 50-paise coins is:

- (A) 36
- (B) 48
- (C) 72
- (D) 60

SSC Selection Post 2021

Q25. 40 litres of a mixture of milk and water contains 10% water. How much water must be added to make the water 20% in the new mixture?

- (A) 4 litres
- (B) 5 litres
- (C) 6 litres
- (D) 8 litres

SSC CGL Tier-II 2021

Q26. In what ratio must a grocer mix two varieties of pulses costing Rs 15 and Rs 20 per kg respectively so as to get a mixture worth Rs 16.50 per kg?

- (A) 3 : 7
- (B) 5 : 7
- (C) 7 : 3
- (D) 7 : 5

SSC CHSL 2020

Q27. The present ages of A and B are in the ratio 4 : 5. After 5 years, they will be in the ratio 5 : 6. The present age of A is:

- (A) 16 years
- (B) 20 years
- (C) 24 years
- (D) 25 years

SSC MTS 2022

Q28. The ratio of the number of boys to that of girls in a school of 432 pupils is 5 : 4. When some new boys and girls are admitted, the number of boys increases by 12 and the ratio of the boys to girls changes to 7 : 6. The number of new girls admitted is:

- (A) 12
- (B) 14
- (C) 20
- (D) 24

SSC CGL Tier-I 2023

Q29. An alloy contains copper, zinc, and nickel in the ratio 5 : 3 : 2. The quantity of nickel (in kg) that must be added to 100 kg of this alloy to have the new ratio 5 : 3 : 3 is:

- (A) 8
- (B) 10
- (C) 12
- (D) 15

SSC CPO 2023

Q30. Two vessels A and B contain acid and water in the ratio 4 : 3 and 5 : 3 respectively. Then the ratio in which these mixtures to be mixed to obtain a new mixture in vessel C containing acid and water in the ratio 3 : 2 is:

- (A) 5 : 8
- (B) 7 : 8
- (C) 7 : 5
- (D) 4 : 7

SSC CGL Tier-II 2020

Q31. If $A : B = 2 : 3$, $B : C = 4 : 5$ and $C : D = 6 : 7$, then $A : B : C : D$ is:

- (A) $16 : 24 : 30 : 35$
- (B) $16 : 22 : 30 : 35$
- (C) $12 : 24 : 30 : 35$
- (D) $16 : 24 : 35 : 30$

SSC CHSL 2021

Q32. The duplicate ratio of $3 : 4$ is:

- (A) $27 : 64$
- (B) $6 : 8$
- (C) $9 : 16$
- (D) $1 : 2$

SSC MTS 2019

Q33. Two numbers are in the ratio $3 : 5$. If 9 is subtracted from each, the new numbers are in the ratio $12 : 23$. The smaller number is:

- (A) 27
- (B) 33
- (C) 49
- (D) 55

SSC CGL Tier-I 2021

Q34. 10 years ago, the average age of a family of 4 members was 24 years. Two children having been born (with age difference of 2 years), the present average age of the family is the same. The present age of the youngest child is:

- (A) 1 year
- (B) 2 years
- (C) 3 years
- (D) 5 years

SSC CPO 2021

Q35. A batsman has a certain average of runs for 11 innings. In the 12th inning, he makes a score of 90 runs, thereby increasing his average by 5. His average after the 12th inning is:

- (A) 30
- (B) 35
- (C) 40
- (D) 45

SSC CGL 2022

Q36. The average score of mathematics of a class of 30 students is 70. When the top two scorers' marks are removed, the average drops to 68.5. If one of the top scorers got 95 marks, the marks of the other top scorer is:

- (A) 85
- (B) 87
- (C) 88
- (D) 90

SSC CHSL 2020

Q37. The average salary of all the workers in a workshop is Rs 8000. The average salary of 7 technicians is Rs 12000 and the average salary of the rest is Rs 6000. The total number of workers in the workshop is:

- (A) 20
- (B) 21
- (C) 22
- (D) 24

SSC Selection Post 2021

Q38. If the average of 6 consecutive even numbers is 25, the difference between the largest and the smallest number is:

- (A) 8
- (B) 10
- (C) 12
- (D) 14

SSC MTS 2020

Q39. If 30% of A is equal to 0.25 of B, which is equal to $\frac{1}{5}$ of C, then A : B : C is:

- (A) 10 : 12 : 15
- (B) 15 : 12 : 10
- (C) 10 : 15 : 12
- (D) 12 : 15 : 10

SSC CGL 2019

Q40. The sum of three numbers is 98. If the ratio of the first to the second is 2 : 3 and that of the second to the third is 5 : 8, then the second number is:

- (A) 20
- (B) 30
- (C) 48
- (D) 58

SSC CPO 2022

Q41. Rs 1050 are divided among A, B and C in such a way that the share of A is $\frac{2}{5}$ of the combined share of B and C. A will get:

- (A) Rs 200
- (B) Rs 300
- (C) Rs 320
- (D) Rs 420

SSC CHSL 2021

Q42. A mixture contains alcohol and water in the ratio 4 : 3. If 5 litres of water is added to the mixture, the ratio becomes 4 : 5. Find the quantity of alcohol in the given mixture.

- (A) 10 litres
- (B) 12 litres
- (C) 15 litres
- (D) 8 litres

SSC CGL Tier-II 2020

Q43. Find two numbers such that their mean proportional is 18 and the third proportional to them is 144.

- (A) 6 and 42
- (B) 9 and 36
- (C) 3 and 18
- (D) 6 and 54

SSC MTS 2021

Q44. What must be added to each term of the ratio 7 : 11, so as to make it equal to 3 : 4?

- (A) 3
- (B) 4
- (C) 5
- (D) 7

SSC Selection Post 2022

Q45. Three containers have their volumes in the ratio 3 : 4 : 5. They are full of mixtures of milk and water. The mixtures contain milk and water in the ratio of (4:1), (3:1) and (5:2) respectively. The contents of all these three containers are poured into a fourth container. The ratio of milk and water in the fourth container is:

- (A) 4 : 1
- (B) 151 : 59
- (C) 157 : 53
- (D) 5 : 2

SSC CGL Tier-II 2021

Q46. A sum of Rs 86,700 is to be divided among A, B and C in such a manner that for every rupee that A gets, B gets 90 paise and for every rupee that B gets, C gets 110 paise. B's share will be:

- (A) Rs 26,000
- (B) Rs 27,000
- (C) Rs 28,000
- (D) Rs 28,090

SSC CPO 2020

Q47. If a, b, c, d are in continuous proportion, then $(a - b) / (c - d)$ is equal to:

- (A) a / c
- (B) b / c
- (C) c / a
- (D) a / d

SSC CHSL 2022

Q48. The average weight of a class of 50 students is 48.6 kg. If the average weight of the 20 boys is 54 kg, then the average weight of the girls in the class is:

- (A) 40 kg
- (B) 42 kg
- (C) 44 kg
- (D) 45 kg

SSC CGL Tier-I 2022

Q49. A person covers half of his journey at 40 km/hr, one-fourth at 20 km/hr and the rest at 10 km/hr. His average speed for the whole journey is:

- (A) 20 km/hr
- (B) 22.8 km/hr
- (C) 26.6 km/hr
- (D) 30 km/hr

SSC CGL 2019

Q50. There are 2 classes A and B having 25 and 30 students respectively. In class A the highest score is 21 and lowest is 17. In class B the highest score is 30 and lowest is 22. Four students are shifted from A to B. What happens to the average of class A?

- (A) Definite increase
- (B) Definite decrease
- (C) Remains constant
- (D) Cannot be determined

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